EVALUATING LIMITS

Matching a known limit

The factor 3/3 equals 1. Its job is to make the denominator equal to the argument of the sine, so the special limit applies exactly.

01 ARGUMENT DOES NOT MATCH — BEFORE

$$\lim_{x \to 0} \frac{\sin 3x}{x}$$

↓ `Multiply by 3/3` — a form of 1: the form changes, not the value

02 DENOMINATOR MATCHES THE ARGUMENT

$$3 \cdot \lim_{x \to 0} \frac{\sin 3x}{3x}$$

↓ `Substitute u = 3x` — as x → 0, u → 0

03 STANDARD FORM

$$3 \cdot \lim_{u \to 0} \frac{\sin u}{u}$$

↓ `Use the special limit` — sin u / u → 1

04 VALUE — AFTER

$$3 \cdot 1 = 3$$

▼ WHY NOT 1?